

Knowledge Base & Company Memory

Employees ask questions in plain language and get answers grounded in TRC's own documents — permissioned by group, cited by source, corrected under admin review.

A shared knowledge layer for the editorial platform: admins upload company documents — policies, playbooks, project histories, some running past a million characters — and the system automatically extracts and indexes them so a new employee can ask a question instead of searching folders or interrupting a colleague. Documents are visible only to the Knowledge Groups they're assigned to, so sensitive material stays restricted without a separate access system per document. This build (Phase 1) ships automatic ingestion of any document size, group-based permissions enforced at query time, source citations on every answer, and an admin-reviewed correction loop; a later phase (§4) adds cross-document relationship reasoning for “how do these connect” questions, once real usage shows plain search-and-cite isn't enough on its own.

STORIES (NOW)	ROLES TOUCHED	EST. EFFORT (NOW)	MAPPED FOR LATER
23 · US-097–127	user, admin	95 hours	8 stories, §4

0. SCOPE & SEQUENCING

This build is Phase 1 only: automatic ingestion of any document size, group-based permissions enforced at query time, cited answers, and an admin-reviewed correction log — specified to full acceptance-criteria depth below. Left deliberately unscoped is a cross-document relationship layer (a lightweight fact graph connecting entities across documents) for questions like “how does this project relate to that client” — plain search-and-cite handles the large majority of onboarding questions on its own, and the added complexity of a relationship graph is only worth paying for once real usage shows it's genuinely needed. §4 keeps that mapped at the same story-ID numbering so nothing has to be re-specified from scratch later.

What's genuinely new here vs. every existing module:

Every existing AI workflow in this app answers from content that fits in a single request — a handful of admin-managed prompts, one transcript, one negotiation. This is the first module built to read from a document library that can grow indefinitely, where any single document can run past a million characters — far larger than any one request to the AI can hold. That's why this brief specifies a search-and-retrieve step (find the relevant passages first, then answer from those) instead of simply handing the AI the whole document, and why it's the first module with a human-reviewed step before a user-submitted fact is allowed to change what every other employee is told.

After launch — who does what:

Uploading, replacing, or removing a document (US-101, US-105, US-106) is an ordinary admin action once this ships — no engineering time, no redeploy, no code change, exactly like adding a category or a user today. Engineering involvement is only needed again for: a genuinely new file type requiring special handling (a spreadsheet, or a scanned/handwritten document needing OCR), a change to how permission groups work, or building the Phase 2 relationship layer in §4. Day-to-day content growth is entirely self-serve for TRC's admins.

1. ROLES

ROLE	ACCESS
user	Existing generic role, gated by a new <code>can_access_knowledge_base</code> flag — same pattern as every other module. Sees only the Knowledge Groups they've been added to; asks questions, sees cited answers, views their own question history, and can flag an answer as wrong or outdated.
admin	Unrestricted, as everywhere else in the app. Uploads, replaces, and removes documents; creates Knowledge Groups and assigns both documents and users to them; reviews and approves/rejects submitted corrections before they affect future answers; sees per-document status, usage/cost analytics, and the full question log.

KEY TECHNICAL DECISIONS

Search-and-retrieve, not “read the whole document.”

A document here can run past a million characters — many times larger than any single request to the AI can hold, even with the largest context windows and caching. So the system breaks each document into indexed passages at upload time and, per question, retrieves only the passages likely to contain the answer — the same principle as a search engine returning the relevant pages rather than reading the entire internet per query. This is genuinely new infrastructure for this app (a search/vector index), not an extension of the “paste a few prompts into context” pattern every other module uses.

Groups, not a permission list per document.

Access is granted at the Knowledge Group level (US-098/US-099) rather than a per-document, per-user list — a document is assigned to one or more groups once, a user is added to a group once, and every document later placed in that group is automatically visible to every member. This keeps admin work bounded as the library grows into the hundreds of documents, instead of scaling with (documents × users).

Corrections are reviewed, not applied instantly.

Unlike a personal coaching tool where a user only corrects facts about their own session, this knowledge base is shared company-wide — a wrong or bad-faith correction from one employee would silently change what every other employee is told. So a submitted correction (US-112) sits in an admin queue until approved (US-113) before it's folded into future answers, rather than taking effect immediately.

2. USER STORIES — SUMMARY (THIS BUILD)

ID	AREA	STORY
FOUNDATION — ACCESS, GROUPS & DATA MODEL		
US-097	ACCESS	Module access is gated by a dedicated permission flag, same pattern as every other module
US-098	GROUPS	Admin creates and names Knowledge Groups to organize document access
US-099	GROUPS	Admin adds and removes users from one or more Knowledge Groups
US-100	DATA	System stores each document as its own record with status, assigned group(s), and version history
INGESTION		
US-101	UPLOAD	Admin uploads a document and assigns it to one or more Knowledge Groups
US-102	INGESTION	System automatically extracts and indexes any document, regardless of size, with no manual step
US-103	EDGE	A document with no extractable text is flagged for attention instead of silently indexing nothing
US-104	INGESTION	System generates a short summary of each document once indexing completes
US-105	UPDATE	Admin replaces an existing document and the system re-indexes it, retiring the old content from future answers
US-106	REMOVE	Admin deletes a document; it stops appearing in future answers immediately, without erasing past chat history that cited it
ASK & ANSWER		
US-107	QUERY	User asks a question in plain language and gets an answer grounded only in their permitted documents
US-108	CITATIONS	Every answer names the document(s) it was drawn from
US-109	EDGE	When no permitted document contains the answer, the system says so rather than guessing
US-110	HISTORY	User can revisit their own past questions and answers
US-111	EDGE	A user is blocked, before any AI call, from asking anything once their access or token limit doesn't allow it
CORRECTIONS		
US-112	CORRECTIONS	User flags an answer as wrong or outdated and states the correct fact
US-113	REVIEW	Admin approves or rejects a submitted correction before it can affect future answers
ADMIN OVERSIGHT		

US-114	ADMIN	Admin sees every document's status, group assignment, size and last-updated date in one table
US-115	ANALYTICS	Admin sees usage and cost for the Knowledge Base module alongside every other module's analytics
US-116	ANALYTICS	Admin can review the full log of questions asked and answers given, for quality control
CROSS-CUTTING		
US-125	EDGE	A document assigned to zero groups is invisible to every non-admin user by default
US-126	EDGE	Deactivating a user immediately revokes their access to every Knowledge Group they were in
US-127	SCALE	A single document of any realistic size — including well past a million characters — is fully searchable and citable, never partially ignored

3. STORY DETAIL & ACCEPTANCE CRITERIA

US-097 | ACCESS

As a user, I can only reach the Knowledge Base if it has been enabled for my account

ACCEPTANCE CRITERIA

- New permission flag (can_access_knowledge_base) on profiles and invitations, following the exact pattern used for every other gated module (Business Cases, Editorial Briefs, Meeting Preparation, Sales Negotiation Coach, Interview Letter Generator)
- Admin grants/revokes access through the existing invite and edit-user flows — no new admin surface needed for this alone
- A user without the flag cannot reach the module via direct URL — same enforcement point as the other gated modules
- Sidebar entry only appears for users with the flag

US-098 | GROUPS

As an admin, I can create and name Knowledge Groups to organize who can see what

ACCEPTANCE CRITERIA

- Admin creates a Knowledge Group with a name and short description (e.g. “All Staff”, “Finance Team”, “Leadership”)
- A group can be renamed or archived later without breaking documents or user assignments already attached to it
- No limit on the number of groups; a document or a user may belong to more than one
- Deleting a group requires confirmation and un-assigns it from every document and user — it does not delete the documents or users themselves

US-099 | GROUPS

As an admin, I can add or remove users from one or more Knowledge Groups

ACCEPTANCE CRITERIA

- Group membership is managed from a simple picker on the group's page or the user's own profile — either surface updates the same underlying membership records
- A user with no group memberships can still use the module but sees zero documents until assigned somewhere
- Removing a user from a group takes effect on their very next question — no stale access from a cached answer
- Admins can see, for any group, the full list of members and assigned documents at a glance

US-100 | DATA

As the system, I store each uploaded document as its own record so status, access and history can all be tracked independently

ACCEPTANCE CRITERIA

- Each document has: title, original filename, uploader, upload date, current status (processing / indexed / failed / needs attention), and its assigned Knowledge Group(s)
- Replacing a document's content creates a new version rather than silently overwriting the old one, mirroring this app's existing prompt-versioning pattern
- The original uploaded file is retained in storage for download/audit, separate from the extracted, indexed text used to answer questions
- Deleting a document is a soft state change (immediately excluded from answers) — see US-106 for what's retained

US-101 | UPLOAD

As an admin, I can upload a document and assign it to one or more Knowledge Groups

ACCEPTANCE CRITERIA

- Accepts common document formats (PDF, Word, plain text) of any realistic company-document length, including a single file well past a million characters
- A document cannot be saved without at least one Knowledge Group selected, or an explicit “visible to all staff” choice — never silently unassigned (see US-125's fail-closed default for anything that slips through)
- Upload succeeds and the document enters “processing” status immediately — indexing happens in the background, the admin isn't blocked waiting for it
- Admin sees a plain confirmation when indexing finishes (or fails) without needing to refresh manually

US-102 | INGESTION

As the system, I automatically extract and index any uploaded document, regardless of size, with no manual step required

ACCEPTANCE CRITERIA

- Text is extracted automatically for any born-digital PDF or Word document — no admin action beyond the upload itself
- The extracted text is split into indexed, searchable passages sized for accurate retrieval, preserving section/heading boundaries where possible so a fact isn't cut away from the context it depends on
- Indexing a very large document (500k+ characters) completes reliably in the background without timing out or requiring the admin to keep the page open
- Re-running indexing on the same document is safe and produces the same result (idempotent), so a retry after a transient failure never produces duplicate content

US-103 | EDGE

As an admin, I'm told when a document can't be indexed instead of it silently producing no results

ACCEPTANCE CRITERIA

- A document with no extractable text (e.g. a scanned image-only PDF) is marked “needs attention” rather than “indexed,” and is excluded from answers until resolved
- The admin sees a plain-language reason (e.g. “no readable text found — this may be a scanned document”)
- The admin can replace the file with a text-readable version (US-105) to resolve it; no separate OCR pipeline is promised in this phase
- A document stuck in “processing” past a reasonable time is surfaced to the admin rather than left silently pending forever

US-104 | INGESTION

As the system, I generate a short summary of each document once indexing completes, so admins and answers can reference what a document is about at a glance

ACCEPTANCE CRITERIA

- Summary is generated automatically, once, right after successful indexing — no separate admin action
- Shown on the document's row in the admin document list (US-114)
- Regenerated automatically whenever the document is replaced (US-105)
- Not shown to end users directly — it's retrieval/admin tooling, not part of the answer itself

US-105 | UPDATE

As an admin, I can replace an existing document's content and have the system re-index it

ACCEPTANCE CRITERIA

- Replacing a document keeps its title, group assignments, and question history intact — only the underlying content and its index change
- The old version's content stops being used in new answers as soon as the replacement finishes indexing; there's no window where both old and new content answer side by side
- Previous versions remain visible in the document's history for audit, mirroring the app's existing prompt-version-history pattern
- Same size/format rules as a fresh upload (US-101) apply to a replacement

US-106 | REMOVE

As an admin, I can delete a document, with confirmation, and it stops appearing in any future answer immediately

ACCEPTANCE CRITERIA

- Deleting requires a confirmation step, same pattern as deleting a user or a category elsewhere in the app
- The document is immediately excluded from retrieval for every user, with no propagation delay
- A user's past question history that cited the deleted document keeps showing the answer they already received — deleting a document doesn't rewrite history, only future answers (mirrors US-011's "history kept" rule for deleted users)
- The original file and its indexed content are retained for a defined retention window (not permanent, not instantly purged) in case of an accidental deletion — exact window to be confirmed with TRC

US-107 | QUERY

As a user, I can ask a question in plain language and get an answer grounded only in documents from my permitted Knowledge Groups

ACCEPTANCE CRITERIA

- Single chat-style input, no need to pick a category or a specific document first
- Retrieval and the permission filter run together — a document outside every group the user belongs to is never even considered, not merely hidden after the fact
- Answer streams back the same way research and other AI outputs already do in this app
- Token-limit and access checks happen before any AI call runs, never after — same rule as every other module (US-006's pattern)

US-108 | CITATIONS

As a user, I can see which document(s) an answer came from

ACCEPTANCE CRITERIA

- Every answer lists the source document title(s) it drew from, visibly alongside the answer, not buried behind a click
- If permissions allow, the user can open/download the cited source document directly from the citation
- An answer that combines facts from more than one document cites all of them, not just the first
- Citations reference the specific document version that was live when the answer was generated (relevant if that document is later replaced per US-105)

US-109 | EDGE

As a user, I'm told when the knowledge base doesn't have an answer, instead of getting a guess

ACCEPTANCE CRITERIA

- If retrieval finds nothing relevant among the user's permitted documents, the system says so plainly rather than answering from general knowledge
- This applies even if a document the user isn't permitted to see would have contained the answer — the user is never told "no answer" in a way that leaks that such a document exists, and its content is never shown either
- The "no answer found" response still counts toward token usage exactly like any other generation, and is still logged (US-116)

US-110 | HISTORY

As a user, I can revisit my own past questions and answers

ACCEPTANCE CRITERIA

- A simple list of the user's own prior questions, most recent first, each opening to its full answer and citations
- Only the asking user (and admins, per US-116) can see a given question — not shared across users by default
- Deleting a document (US-106) does not remove it from past answers already shown here

US-111 | EDGE

As a user, I'm blocked from asking a question — before any AI cost is spent — the moment my access or budget doesn't allow it

ACCEPTANCE CRITERIA

- A user without the module's permission flag never reaches the question screen (US-097)
- A user who has reached their monthly token limit sees a clear message and cannot submit a question, exactly like the research module's existing token-limit gate (US-006)
- Neither check makes an AI call — the block happens client- and server-side before generation starts

US-112 | CORRECTIONS

As a user, I can flag an answer as wrong or outdated and state what the correct fact actually is

ACCEPTANCE CRITERIA

- Flag control sits directly next to the answer it applies to
- User states the correct fact in their own words; the system also snapshots what the AI originally said, for the admin's review
- Submitting a correction does not change any answer immediately — it enters the admin review queue (US-113)
- User can see the status of their own submitted corrections (pending / approved / rejected)

US-113 | REVIEW

As an admin, I review every submitted correction and approve or reject it before it can affect future answers

ACCEPTANCE CRITERIA

- Admin sees the original AI answer, the user's stated correction, which document (if any) it relates to, and who submitted it
- Approving a correction makes it part of the grounding for every future question that touches that fact, from every user — not just the one who submitted it
- Rejecting a correction discards it; it never affects future answers, and the submitting user isn't automatically renotified with a reason unless the admin adds one
- No AI call is made to apply an approved correction — it's added as trusted context for future questions, the same “fold every approved entry into every future run” pattern already used for Sales Coach corrections

US-114 | ADMIN

As an admin, I see every document's status, group assignment, size, and last-updated date in one table

ACCEPTANCE CRITERIA

- Table columns: title, status, assigned Knowledge Group(s), file size / approximate length, uploader, last updated
- Sortable/searchable, same UX pattern as the existing admin user and category tables
- A document in "needs attention" status (US-103) is visually flagged so it doesn't get missed
- Clicking a row opens that document's detail (versions, summary, which groups/users can see it)

US-115 | ANALYTICS

As an admin, I see usage and cost for the Knowledge Base module alongside every other module's analytics

ACCEPTANCE CRITERIA

- Every question is logged as its own usage_events row (workflow: knowledge_base_query), following the existing ledger pattern used by every other AI workflow in this app
- Appears in the existing analytics dashboard's per-workflow breakdown and per-user cost table — no separate analytics page needed
- Indexing a document (extraction + embedding) is also logged, so document-processing cost is visible and not hidden inside "free" admin actions

US-116 | ANALYTICS

As an admin, I can review the full log of questions asked and answers given, for quality control

ACCEPTANCE CRITERIA

- Log includes the asking user, the question, the answer, citations, and timestamp
- Filterable by user and by date range, same UX conventions as the existing admin tables
- Corrections raised against a logged answer (US-112) are visible inline with that entry
- This log is admin-only — a user only ever sees their own history (US-110), never another user's

US-125 | EDGE

As the system, I never show a document to a user unless it's explicitly assigned to a group they belong to

ACCEPTANCE CRITERIA

- A document with zero assigned Knowledge Groups is invisible to every non-admin user by default — fail closed, not fail open
- A document's indexing status and its visibility are independent — "indexed" never implies "visible to someone"
- Admins are warned if they try to leave a freshly uploaded document with no group assigned (see US-101)

US-126 | EDGE

As the system, deactivating a user immediately removes their access to every Knowledge Group they were in

ACCEPTANCE CRITERIA

- Mirrors the app-wide “deactivation = instant logout” rule (US-010) — a deactivated user's group memberships stop granting access on their very next request, not merely at next login
- Reactivating a user restores their prior group memberships as they were, rather than requiring an admin to reassign them from scratch
- Deleting a user (not just deactivating) follows the existing “history kept” rule — their past questions stay in the log (US-116) attributed to them, or to “Deleted user” if the profile itself is gone, mirroring US-011/US-016a

US-127 | SCALE

As a user, I get a complete, accurate answer even when it depends on a single document far larger than any one request to the AI could hold

ACCEPTANCE CRITERIA

- A document of any realistic size (tested against at least one document exceeding a million characters) is fully searchable — no silent truncation of “the parts that didn't fit”
- An answer that depends on facts from two different sections of the same large document (not just one passage) still finds and cites both
- Ingestion time for a very large document is proportional to its size but never blocks the admin's other work, and never fails purely due to length

4. MAPPED FOR LATER — PHASE 2 MENU (NOT IN THIS BUILD)

Not committed, not estimated, not sequenced against Phase 1 — a menu to pick from once real usage of the Phase 1 build actually calls for it. IDs are reserved now (US-117–124) so nothing here has to be re-numbered or re-specified from scratch when it's picked up.

US-117 GRAPH

At ingestion, the system additionally extracts named entities and relationships from each document into a lightweight fact graph, so facts that span documents can be connected, not just found by keyword/similarity.

US-118 GRAPH

A user's question that touches more than one document ("how does X relate to Y") retrieves connected graph facts alongside document passages, not vector search alone.

US-119 GRAPH

Admin can view a document's extracted entities/relationships and correct or remove an obviously wrong one.

US-120 EXPANSION

Support for additional file types requiring special handling (spreadsheets, scanned/handwritten documents via OCR).

US-121 EXPANSION

Documents can be organized into a folder/topic hierarchy within a Knowledge Group, for browsing rather than only asking.

US-122 NOTIFICATIONS

Users are notified when a document relevant to a saved question is added or updated.

US-123 EXPORT

Admin can export the full corrections log and document library metadata for external audit.

US-124 PERSONALIZATION

Per-user memory of previously stated preferences/context (beyond shared corrections), so the assistant can tailor phrasing to a specific employee's role without changing the shared facts everyone else sees.